Will It Stick? A Feasibility Study on Fan-powered cars to Drive Upside Down

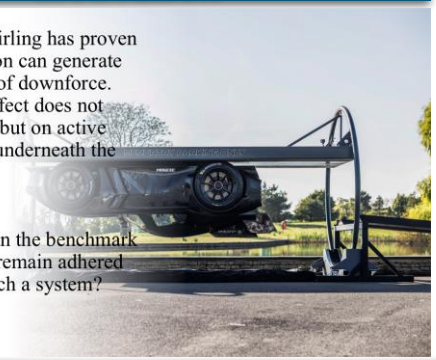
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Motivation

• The McMurtry Spéirling has proven that fan-driven suction can generate extraordinary levels of downforce. Unlike wings, this effect does not depend on the speed but on active pressure depression underneath the car.

This Study asks: Can the benchmark DrivAer model also remain adhered upside down with such a system?



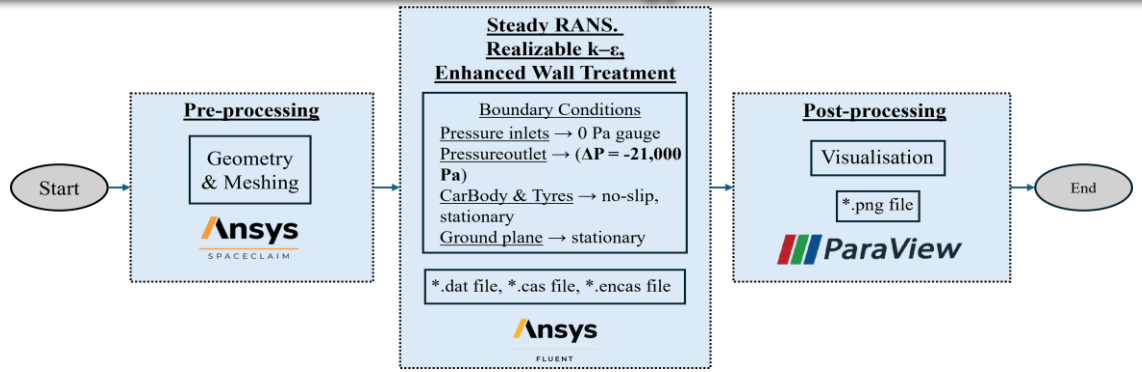
Aim and objectives

Aim: Determine if a fan-suction system can generate sufficient downforce to keep a modified DrivAer adhered upside down.

Objectives:

- Modify the DrivAer underbody with restrictor walls and twin fans.
- Perform CFD to evaluate downforce, mass flow, and leakage behaviour.
- Assess whether the system can achieve inverted adhesion.

Methodology



Summary

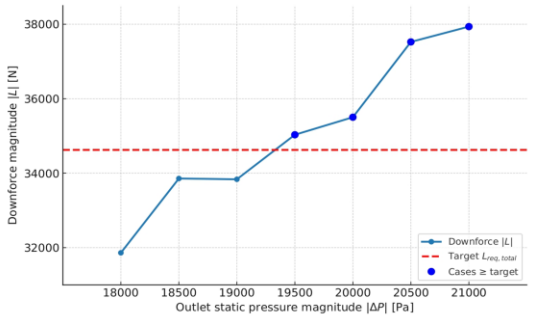
•ΔP = –21 kPa was identified as the design point, giving ~38 kN downforce (≈9.6% margin).

•Optimal configuration: ride height ≈ 20 mm, skirt clearance ≈ 10 mm.

•Increased ride or skirt height leads to leakage and reduced suction efficiency.

Results and Discussions

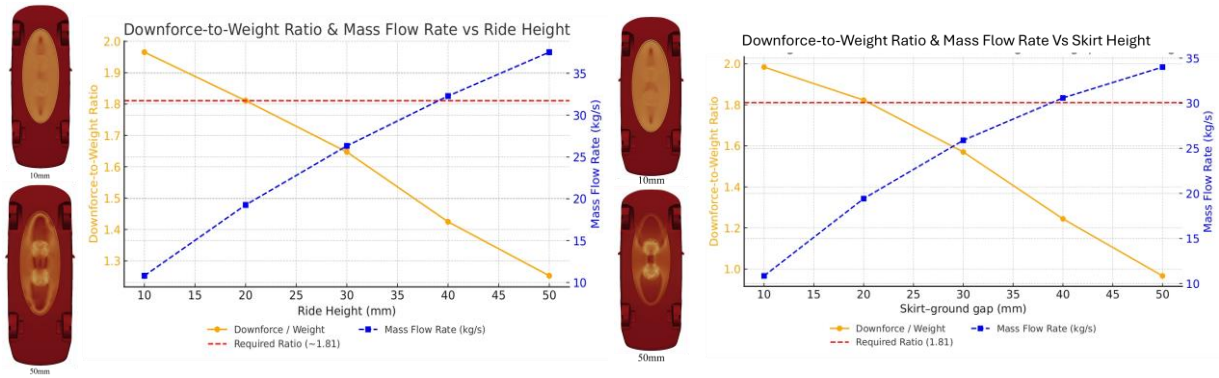
Downforce Vs Suction System



Fan System Requirements for the Experiment

Parameter	Value
Required adhesion load (SF = 1.8, 1950.00 kg)	34.62 kN
Downforce at $\Delta P = -21.00$ kPa	37.94 kN (9.60 % margin)
Mass flow rate	10.85 kg s^{-1} (≈ 9.04 m^3/s)
Per-fan duty point	$Q = 4.52$ m^3/s at $\Delta P = 21.00$ kPa
Tip speed / RPM (D = 0.46 m to 0.48 m)	≈ 187.00 m s^{-1}, 7440.00 rpm to 7770.00 rpm
Electrical input (two fans)	≈ 278.00 kW
Energy for 120.00 s inverted run	≈ 9.30 kW h
Fan system mass (two modules)	270.00 kg to 320.00 kg (conservative)
Vehicle mass with fans	≈ 2000.00 kg

Downforce increases with suction level. The required load was set as **vehicle weight + fan system**, multiplied by a **1.8** safety factor. At –21 kPa, the system produced ~37.9 kN, surpassing the adhesion requirement and chosen as the design point.



Downforce-to-weight ratio decreases with increasing ride and skirt height, while mass flow demand rises. Pressure contours show that higher clearances allow leakage around the skirts, weakening the suction pocket. Effective sealing is therefore critical to sustaining inverted adhesion.

MSc in Computational Fluid Dynamics